

Paul Mihalitsas

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Education

University of Toronto 2025 – 2029

Bachelor of Science – Mathematical Applications in Finance

Relevant coursework: Calculus(MAT137), Linear Algebra, Probability & Statistics, Financial Mathematics

Projects

Volatility Forecasting & Iron Condor Strategy Generator 2025 – Present

Python | XGBoost | Random Forest | Options Pricing | Time Series Analysis

- Built a two-component system combining volatility forecasting with automated iron condor options strategy construction, directly applicable to derivatives risk management.
- Trained XGBoost and Random Forest models on historical market data to predict 5-day forward volatility; RandomForest achieved R^2 of 0.696 and MAE of 0.025.
- Engineered features including rolling volatility windows, Bollinger Bands, ATR, and regime-change clustering to capture market microstructure dynamics.
- Automated iron condor strike selection using delta targets, spread-width optimization, and risk/reward calculations (max profit, max loss, break-even points).
- Applied options pricing theory (Black-Scholes sensitivities) to assess probability of profit and generate actionable trade management rules.

News-Driven Stock Movement Classifier (XGBoost) 2025 – Present

Python | XGBoost | Gradient Boosting | GridSearchCV | Sentiment Analysis

- Developed a binary classifier to predict significant stock price movements (>2% absolute return) from news sentiment and market features, achieving 91.7% accuracy after hyperparameter tuning.
- Compared baseline Gradient Boosting (87.4%) against XGBoost (89.2%) and tuned XGBoost via GridSearchCV (91.7%) to identify the optimal model configuration.
- Handled class imbalance (68%/32% split) and implemented stratified train/test splitting to ensure representative model evaluation.
- Designed a practical trading signal framework mapping model predictions to volatility-based strategy recommendations (straddles vs. covered calls) with risk management protocols.

NHL Goal-Scoring Prediction System (Tim's Challenge) Dec 2025 – Present

Python | REST APIs | Probability Modelling | Data Analytics

- Built a Python pipeline pulling live data from the NHL Stats API and applying a mathematical reward model to rank likely goal scorers each game day.
- Achieved 74% accuracy (28/38) on player selection, placing in the top 3% of Tim Hortons app participants with a peak score of 0.54.

Neural Networks for Medical Classification March 2026

- Trained a neural network on the Wisconsin Diagnostic dataset achieving 99% accuracy in breast cancer classification.
- Built a CNN for chest X-ray pneumonia detection, reaching 100% accuracy on unseen test data through careful preprocessing and architecture design.

Technical Skills

Languages: Python

ML / Data: XGBoost, Gradient Boosting, Random Forest, Neural Networks (TensorFlow/Keras), scikit-learn, Pandas, NumPy, Excel

Finance / Quantitative: Options pricing, volatility modelling, time series analysis, risk/reward frameworks, financial data analysis

Soft Skills: Communication, teamwork, analytical problem-solving, self-directed research